HealthConnect REFINED Layer – Data Model Documentation (2025)

Prepared for: HealthConnect Data Warehouse Project
Version: 1.0 | Environment: REFINED Layer | Schema: refined_dw | Author: Data Engineering Team

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Project Overview

The **HealthConnect REFINED Layer** is the third and final layer of the HealthConnect Data Warehouse pipeline, stored under the **refined_dw** schema. It is derived from the CLEANSED layer after applying business rules, transformations, and validations. This layer provides **curated, analytics-ready data** for reporting and downstream applications. The **load_date** captures initial ingestion, and **last_updated_date** tracks incremental updates or late-arriving data changes.

Table: refined_patients

Column	Type	Constraints
patient_id	INT	
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
gender	CHAR(1)	
date_of_birth	DATE	
state_code	CHAR(2)	
city	VARCHAR(50)	
phone	VARCHAR(15)	
load_date	DATE	
last_updated_date	DATE	

Table: refined_providers

Column	Type	Constraints
provider_id	INT	
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
specialty	VARCHAR(50)	
npi	VARCHAR(20)	
load_date	DATE	
last_updated_date	DATE	

Table: refined_payers

Column	Type	Constraints
payer_id	INT	
payer_name	VARCHAR(100)	
load_date	DATE	
last_updated_date	DATE	

Table: refined_encounters

Column	Type	Constraints
encounter_id	INT	
patient_id	INT	
provider_id	INT	
encounter_type	VARCHAR(10)	
encounter_start	DATETIME	
encounter_end	DATETIME	
height_cm	INT	
weight_kg	FLOAT	

systolic_bp	INT	
diastolic_bp	INT	
load_date	DATE	
last_updated_date	DATE	

Table: refined_diagnoses

Column	Type	Constraints
diagnosis_id	INT	
encounter_id	INT	
diagnosis_description	VARCHAR(100)	
is_primary	VARCHAR(10)	
load_date	DATE	
last_updated_date	DATE	

Table: refined_procedures

Column	Type	Constraints
procedure_id	INT	
encounter_id	INT	
procedure_description	VARCHAR(100)	
load_date	DATE	
last_updated_date	DATE	

Table: refined_medications

Column	Type	Constraints
medication_id	INT	
encounter_id	INT	
drug_name	VARCHAR(50)	
route	VARCHAR(20)	
dose	VARCHAR(20)	
frequency	VARCHAR(10)	
days_supply	INT	
load_date	DATE	
last_updated_date	DATE	

Table: refined_claims

Column	Type	Constraints
claim_id	INT	
encounter_id	INT	
payer_id	INT	
admit_date	DATE	
discharge_date	DATE	
total_billed_amount	FLOAT	
total_allowed_amount	FLOAT	
total_paid_amount	FLOAT	
claim_status	VARCHAR(20)	
load_date	DATE	
last_updated_date	DATE	